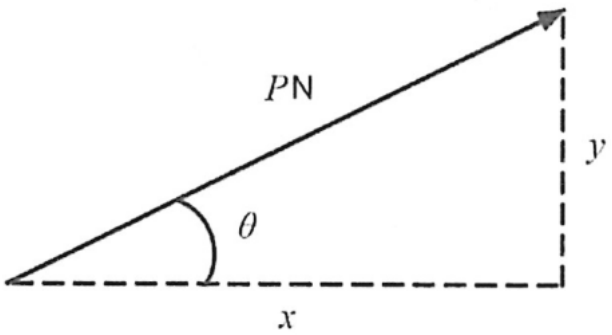
Forces: Equilibrium

Questions 1–16

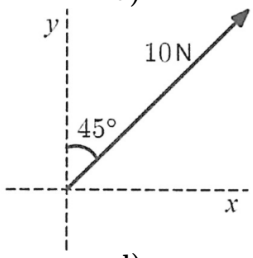
Question 1. Find the horizontal and vertical component x and y of the force P as shown in the diagram below, for each of the given values of P and θ:



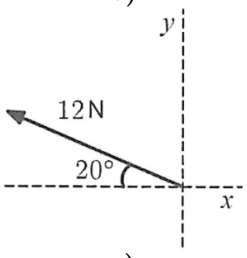
a) $P = 5$ N, $\theta = 30°$
b) $P = 6$ N, $\theta = 45°$
c) $P = 8$ N, $\theta = 40°$
d) $P = 3$ N, $\theta = 22°$

Question 2. Find the horizontal and vertical component of each of the forces shown:

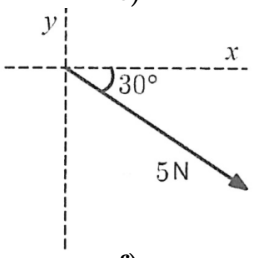
a)



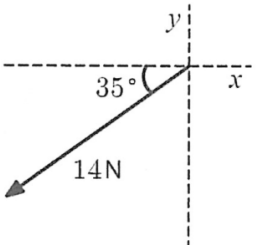
b)



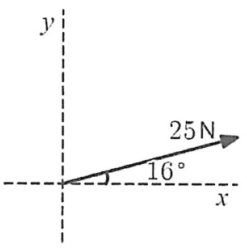
c)



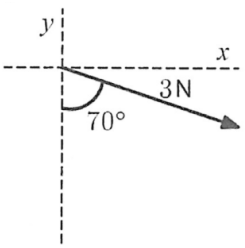
d)



e)

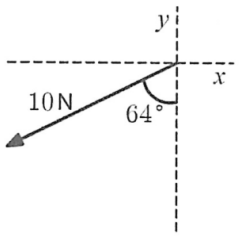


f)

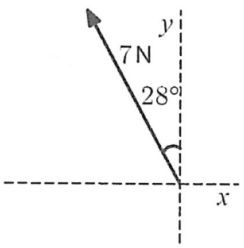


Question 2 (continued). Find the horizontal and vertical component of each of the forces shown:

g)

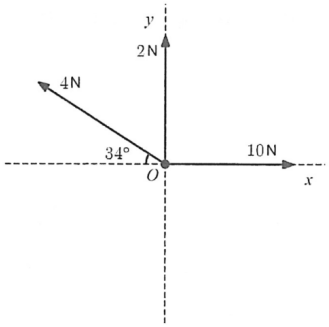


h)

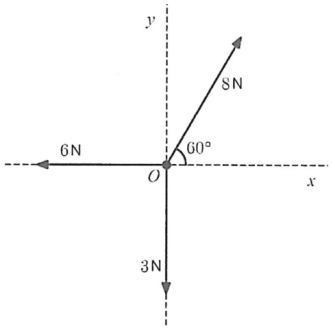


Question 3. For each of the diagrams below, a particle is acted upon by the forces shown. Find the magnitude of the resultant force and the angle the resultant makes with Ox:

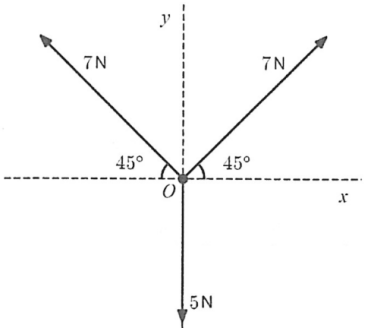
a)



b)

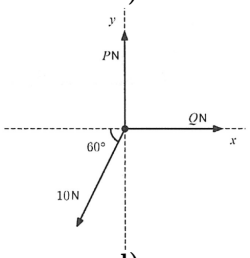


c)

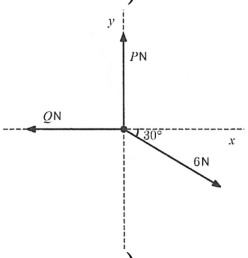


Question 4. In each of the diagrams below, a particle is at rest, under the action of the forces shown.
In each case, by resolving in the x-direction and y-direction, find the magnitude of any unknown forces (labelled P, Q, R) and the size of any unknown angles (labelled θ):

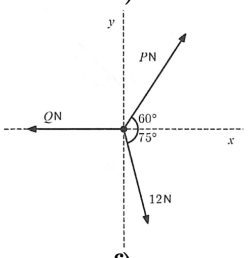
a)



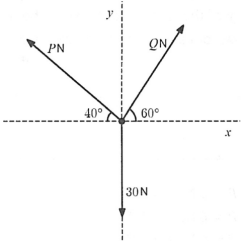
b)



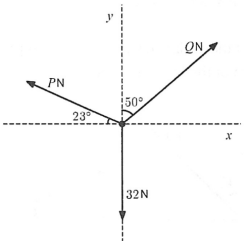
c)



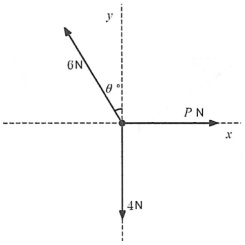
d)



e)

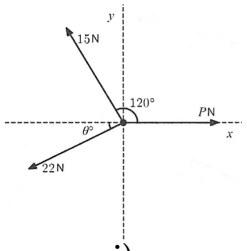


f)

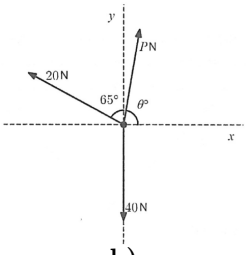


Question 4 (continued). In each of the diagrams below, a particle is at rest, under the action of the forces shown.
In each case, by resolving in the x-direction and y-direction, find the magnitude of any unknown forces (labelled P, Q, R) and the size of any unknown angles (labelled θ):

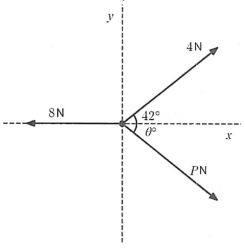
g)



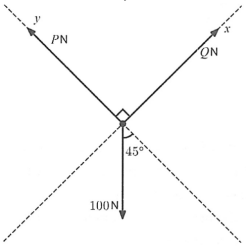
h)



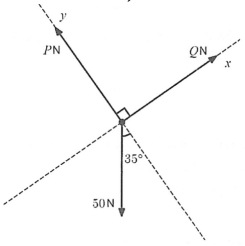
i)



j)

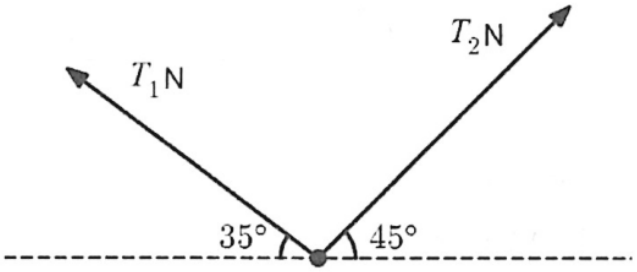


k)

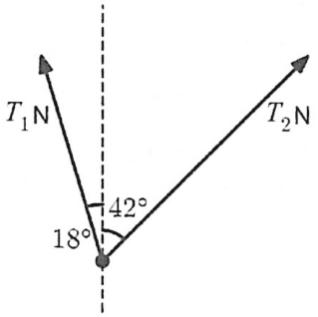


Question 5. The following questions use $g = 9.8$ **ms**$^{-2}$ An object of mass 2 kg is held in equilibrium, suspended by two light, inextensible strings as shown in each of the diagrams. The tensions in the string are given by T_1 N and T_2 N.
In each case, find the value of T_1 and T_2

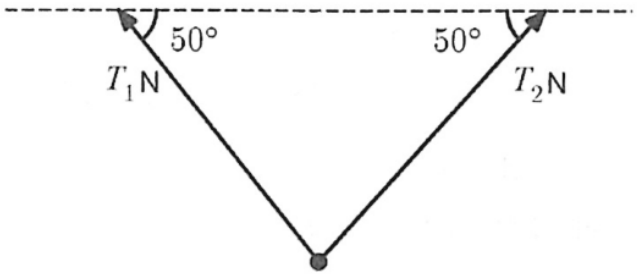
a)



b)

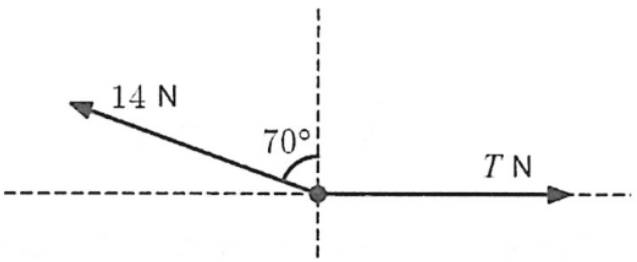


c)

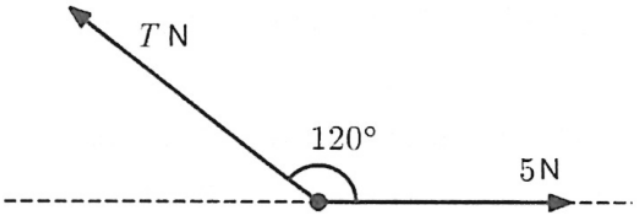


Question 6. A particle of mass m kg is held in equilibrium, suspended by two light, inextensible strings as shown in each of the diagrams. One of the strings has tension T
In each case, find the value of T and m

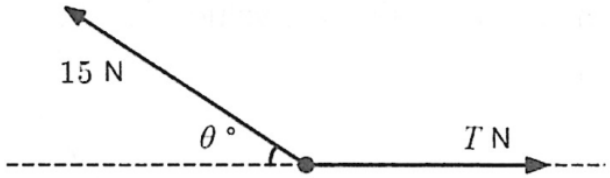
a)



b)



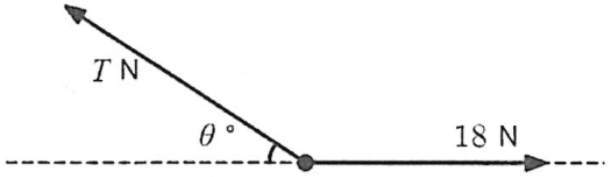
Question 7. A particle of weight 10 N is held in equilibrium, suspended by two light, inextensible strings as shown in the following diagram. The tension in the horizontal string is T N. The second string is pulling with a tension of 15 N at $\theta°$ to the horizontal in the opposite direction to T



Find

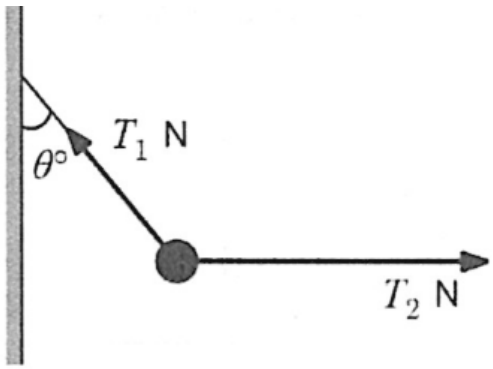
a) the value of θ
b) the value of T

Question 8. A particle of weight 4 N is held in equilibrium, suspended by two light, inextensible strings as shown in the following diagram. The tension in the horizontal string is 18 N. The second string is pulling with a tension of T N at $\theta°$ to the horizontal in the opposite direction to the first string.



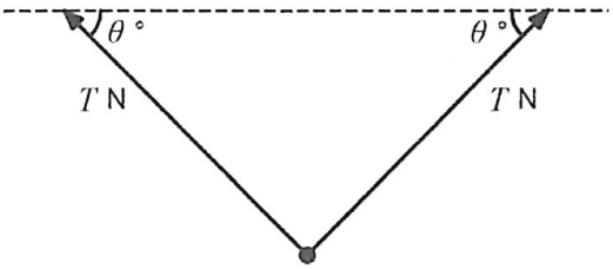
a) Find the value of θ
b) Find the value of T

Question 9. The following questions use $g = 10$ **ms**$^{-2}$ A box of mass 4 kg is suspended by a cable, which makes an angle $\theta°$ with a vertical wall. A horizontal rope keeps the box in equilibrium.



a) Calculate the tension in the cable giving your answer in terms of θ
b) Calculate the tension in the rope giving your answer in terms of θ
c) Find the angle at which the tension in the cable and the rope is the same.
d) Find the range of angles at which the tension in the rope is less than the weight of the box.

Question 10. The following questions use exact values of g A particle of mass m kg is held in equilibrium by two strings, each inclined at $\theta°$ to the horizontal. The tension in each string is T N.

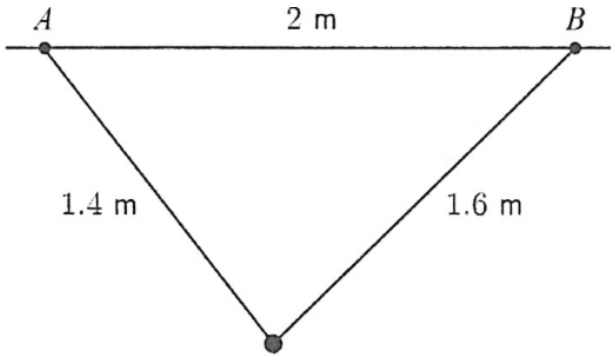


a) Show that

$$T = \frac{mg}{2\sin\theta°}$$

Find T when the particle has weight 5 N and
b) θ is 45°
c) θ is 60°
d) $\theta°$ approaches 90°

Question 11. Two ropes are hung from a ceiling, 2 metres apart and are connected at their other ends. The length of rope A and rope B are 1.4 metres and 1.6 metres respectively.



A crate of mass m kg is attached at the end of both rope A and rope B. The system is in equilibrium. Show that

a) the acute angle α between rope A and the ceiling is given by

$$\cos\alpha = \frac{17}{28}$$

b) the acute angle β between rope B and the ceiling is given by

$$\cos\beta = \frac{23}{32}$$

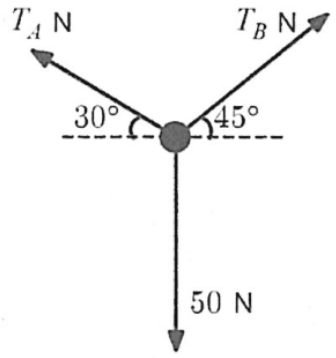
The tensions in rope A and rope B are given by T_A and T_B respectively.

c) Show that $\frac{T_B}{T_A} = \frac{136}{161}$

Given that $T_A = 322$ N,

d) find the exact value of m in terms of g

Question 12. The following questions use Triangles of Forces and $g = 9.8$ **ms**$^{-2}$ An object of weight 50 N is suspended, in equilibrium, by two light inextensible strings A and B which make angles of 30° and 45° with the horizontal as shown in the diagram below:

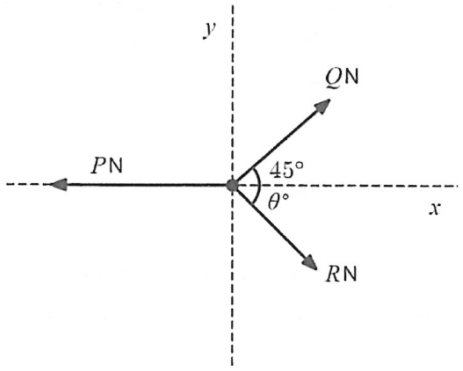


The tension in string A is given by T_A N and the tension in string B is given by T_B N.

a) Sketch the triangle of forces for these forces.

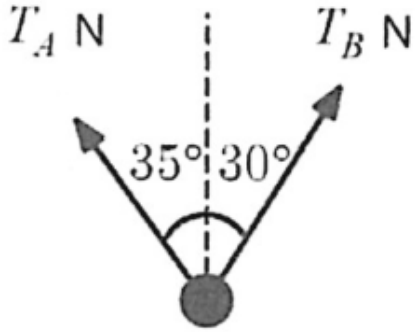
b) Hence determine the value of T_A and T_B

Question 13. The diagram below shows a system of forces in equilibrium:



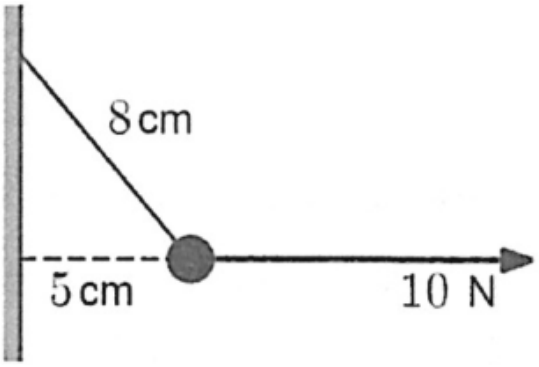
a) Sketch the triangle of forces for these forces.
b) Find the range of possible values of θ for the system to remain in equilibrium.

Question 14. A particle of mass 2 kg is suspended by two light inextensible strings A and B which make an angles of 35° and 30° to the vertical respectively, as shown in the diagram below:



The tension in string A is given by T_A N and the tension in string B is given by T_B N.
a) Sketch the triangle of forces for the forces acting on the object.
b) Hence determine the value of T_A and T_B

Question 15. An object of weight w N is attached to one end of a light inextensible string of length 8 cm. The other end is attached to a vertical wall at the point A. A horizontal force of magnitude 10 N is applied to the object. The particle is in equilibrium with the string taut, where it lies 5 cm from the wall.



a) Sketch a triangle of forces for the forces acting on the object.

Hence find

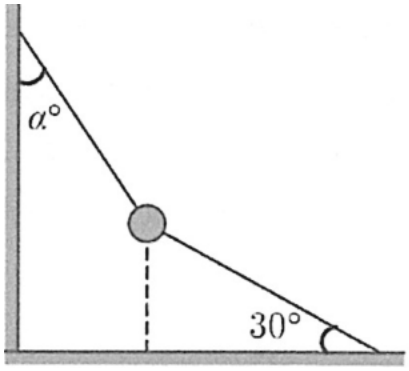
b) the tension in the string.

c) the value of w

d) the mass of the object.

Question 16. The following questions use exact values of g

Prerequisite knowledge for Q16: Compound Angle Formulae A small box B of mass m kg is held in equilibrium by two light strings, one attached to a vertical wall which has a tension of T_W N and one to the horizontal floor which has a tension of T_F N. They make an angle of $\alpha°$ with the wall and 30° with the floor respectively.



a) Sketch a triangle of forces for the forces acting on the box.

Hence or otherwise, given $\alpha = 30°$, show that

b) $T_W = \sqrt{3}mg$

c) $T_F = mg$

Question 16 (continued). Angle α is increased and the angle of the string with the floor becomes $45°$. The system remains in equilibrium.

d) Show that

$$T_W = \frac{mg}{\cos \alpha° - \sin \alpha°}$$